

# Pseudo-Laplacians at Whitney Cross-Caps: Point Interactions on Surfaces Mapped into Three-Space

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## Abstract

On a smooth Riemannian surface, Colin de Verdière showed that the Laplacian restricted to functions vanishing near a point admits a one-parameter family of self-adjoint extensions, the pseudo-Laplacians, whose domains are described by a logarithm and a constant. We show that the same picture holds at the Whitney cross-caps of a surface mapped into  $\mathbb{R}^3$ . These are the generic singularities of such maps, and the induced metric degenerates at them. Each cross-cap carries exactly one logarithmic channel. The leading coefficient is the universal  $-1/(2\pi)$  in the intrinsic distance, or equivalently in the extrinsic radius, and the regular part is Hölder continuous. With  $k$  cross-caps, the extensions of the point-restricted Laplacian are classified by  $U(k)$  in explicit geometric boundary coordinates with a  $2\pi$  Green pairing. The Friedrichs extension is the unique extension with a sub-Markovian semigroup, and the unique one invariant under change of logarithmic reference length. The key step is an arclength coordinate in which the metric is the identity up to  $O(\rho^{1/2})$ , refining Grieser's quasi-isometry of the Whitney umbrella. Examples include Steiner's Roman surface: its six cross-caps carry a  $U(6)$ -family of pseudo-Laplacians, and octahedral symmetry forces its Green matrix to have at most three eigenvalues.

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# 1 Introduction

Let  $X$  be a smooth Riemannian surface and  $x_0 \in X$ . Colin de Verdière [1] observed that the Laplacian on  $C_c^\infty(X \setminus \{x_0\})$  is not essentially self-adjoint. Its deficiency indices are  $(1, 1)$ , and the self-adjoint extensions, the *pseudo-Laplacians*, are singled out by the behaviour

$$f(x) = \lambda \left( \sin \alpha \frac{1}{2\pi} \log r + \cos \alpha \right) + o(1), \quad r = d(x, x_0).$$

This picture underlies point interactions on manifolds. For conical points of a flat surface [2], the logarithm is again the only channel when the cone angle is at most  $2\pi$ ; larger angles add power-law channels.

Generic smooth maps of a compact surface into  $\mathbb{R}^3$  are not immersions. Whitney showed that their only singularities are isolated *cross-caps* [4, 5]. At a cross-cap the differential has rank one, and the induced metric  $g = f^*\delta$  degenerates. This paper asks what happens to point interactions at these points, and gives a complete answer: a cross-cap behaves exactly like a regular point of the surface.

**Main results.** Let  $f : \Sigma \rightarrow \mathbb{R}^3$  be a smooth map of a compact surface, immersive except at finitely many cross-caps; the precise setting is Definition 2.1. Let  $H_F$  be the Friedrichs Laplacian of  $g$ , with Dirichlet condition if  $\Sigma$  has a boundary, and let  $p_1, \dots, p_k$  be the interior cross-caps.

1.  $H_F$  has compact resolvent (Theorem 4.2). Every  $u \in \text{Dom } H_F$  is continuous at each  $p_j$ , and  $\tau u = (u(p_j))_j$  is a graph-bounded surjection onto  $\mathbb{C}^k$  with dense kernel (Theorem 5.1).
2. The restriction  $A = H_F|_{\ker \tau}$  has deficiency indices  $(k, k)$ , and its self-adjoint extensions form a  $U(k)$ -family with explicit Krein resolvent formula (Theorem 5.2).
3. Every  $\varphi \in \text{Dom } A^*$  has the expansion  $\varphi = \ell_j \log d_g(\cdot, p_j) + b_j + O(d_g^\beta)$  at each  $p_j$ , and the Green form is  $2\pi(\langle \ell, b' \rangle - \langle b, \ell' \rangle)$ . In particular the resolvent Green vectors have leading term  $-\frac{1}{2\pi} \log d_g$  (Theorems 6.1 and 6.2).
4.  $H_F$  is the only extension with a sub-Markovian semigroup, and the only one invariant under a common change of logarithmic reference length. Coefficients on the two sheets meeting at a cross-cap agree in the limit (Section 7).

Each cross-cap therefore carries one channel, as a regular point does, and its Whitney invariants enter only through the regular part. The geometric input is an arclength coordinate (Lemma 3.2) in which the metric equals the identity up to  $O(\rho^{1/2})$ . This refines Grieser's quasi-isometry of the Whitney umbrella with the plane [3] from bounded distortion to asymptotic equality, which is what fixes the exact coefficient  $-1/(2\pi)$ .

The theory is local at the cross-caps but needs a global boundary condition. If one instead closes the Laplacian on functions compactly supported in the interior, every regular boundary arc contributes infinitely many deficiency modes (Section 8). The examples in Section 9 are Steiner's Roman surface, a closed surface with six cross-caps, and a ruled surface from an earlier signal model, which also has cross-caps on its boundary.

**Relation to the literature.** The abstract extension theory is Posilicano's [7], and the scalar regularity input is the De Giorgi–Nash–Moser theory in the form of [9]. Colin de Verdière [1] treats regular points of smooth surfaces and Hillairet–Kokotov [2] conical points. Grieser [3] supplies the quasi-isometric coordinates. The contribution here is to carry this circle of ideas to the generic singular points of surfaces in  $\mathbb{R}^3$ , together with the sharpened local geometry that this requires.

## 2 Setting

**Definition 2.1.** Let  $\Sigma$  be a compact connected smooth surface, either closed or with a piecewise smooth Lipschitz boundary, and let  $f : \Sigma \rightarrow \mathbb{R}^3$  be smooth up to the boundary. Assume that  $f$  is an immersion except on a finite set  $Z \cup Z_\partial$ , where  $Z \subset \text{int } \Sigma$  and  $Z_\partial \subset \partial\Sigma$ , and that every point of  $Z \cup Z_\partial$  is a *Whitney cross-cap* of  $f$ , or of a smooth extension of  $f$  near  $\partial\Sigma$ . That is,  $df$  has rank one there, and for a kernel vector  $\eta$  and a complementary vector  $\xi$ ,

$$\det(f_\xi, f_{\xi\eta}, f_{\eta\eta}) \neq 0. \quad (1)$$

At each  $p \in Z_\partial$  we assume that, in some smooth chart of the extension centred at  $p$ ,  $\Sigma$  is a closed sector  $\{\theta_1 \leq \theta \leq \theta_2\}$ . We write  $Z = \{p_1, \dots, p_k\}$ . There are two regimes:

(D)  $\partial\Sigma \neq \emptyset$ , with the Dirichlet condition imposed on the whole boundary;

(N)  $\Sigma$  is closed, in which case  $Z_\partial = \emptyset$ .

The condition (1) is the standard recognition criterion for the cross-cap  $(x, y) \mapsto (x, xy, y^2)$  [6]. A theorem of Whitney [4], see also [5], states that the cross-cap is the only stable singularity of maps from surfaces to  $\mathbb{R}^3$ . Consequently, for a generic map the hypotheses hold with  $Z_\partial = \emptyset$ .

Put  $g = f^*\delta$  on  $\Sigma \setminus (Z \cup Z_\partial)$ . Let  $\mathcal{H} = L^2(\Sigma, dA_g)$  and  $q(u, v) = \int \langle \nabla u, \nabla v \rangle_g dA_g$ . Take as core  $\mathcal{C} = C_c^\infty(\text{int } \Sigma \setminus Z)$  in case (D), and the smooth functions on  $\Sigma$  vanishing near  $Z$  in case (N). The closure  $\mathcal{V}$  of  $\mathcal{C}$  in the norm  $(q(u, u) + \|u\|^2)^{1/2}$  defines a closed form, and we let  $H_F$  be its self-adjoint operator. Scalar products are conjugate-linear in the first slot.

## 3 Local geometry at a cross-cap

**Lemma 3.1** (Adapted form). *After a translation and a rotation of  $\mathbb{R}^3$  and a smooth change of source coordinates centred at a cross-cap,  $f(x, y) = (x, h(x, y))$  with  $h(0) = 0$ ,  $Dh(0) = 0$  and*

$$h_y(x, y) = xv + 2yw + O(R^2), \quad R = (x^2 + y^2)^{1/2},$$

where  $v, w \in \mathbb{R}^2$  are linearly independent. Hence  $cR \leq |h_y| \leq CR$  and  $|h_x| \leq CR$ . In any smooth chart centred at the point,  $g \leq CI$  and  $cR^2 \leq \det g \leq CR^2$ ; in particular  $dA_g \asymp R dx dy$ .

*Proof.* Rotate so that  $f_\xi(0)$  points along the first axis. Use  $x$ , the first component of  $f$ , together with a coordinate  $y$  along the kernel direction. Then  $h_x(0) = 0$  and  $h_y(0) = 0$ . The vectors  $v = h_{xy}(0)$  and  $2w = h_{yy}(0)$  are independent by (1). In these coordinates  $g = \begin{pmatrix} 1+|h_x|^2 & h_x \cdot h_y \\ h_x \cdot h_y & |h_y|^2 \end{pmatrix}$ , whose entries are bounded and whose determinant lies between  $|h_y|^2$  and  $(1 + |h_x|^2)|h_y|^2$ . A smooth change of chart has bounded derivative and inverse, so these bounds persist.  $\square$

**Lemma 3.2** (Arclength coordinate). *In the coordinates of Lemma 3.1 put*

$$V(x, y) = \int_0^y |h_y(x, s)| ds, \quad \Xi(x, y) = (x, V), \quad \rho = |\Xi|.$$

*Then  $\Xi$  is a homeomorphism of a neighbourhood of 0 onto a neighbourhood of 0, and a  $C^1$  diffeomorphism off 0. In the coordinates  $(X, V) = \Xi$ , the metric  $G$ , the density  $m = \sqrt{\det G}$  and the divergence tensor  $a = mG^{-1}$  satisfy*

$$G - I = O(\rho^{1/2}), \quad a - I = O(\rho^{1/2}), \quad m - 1 = O(\rho),$$

*and are bounded and uniformly elliptic. The extrinsic radius  $\rho_e = |f - f(0)|$  satisfies  $\rho_e/\rho = 1 + O(\rho^{1/2})$ .*

*Proof.* For fixed  $x$ ,  $V$  is strictly increasing in  $y$ , and  $cy^2 \leq |V| \leq C(|xy| + y^2)$ . Its inverse is therefore continuous, also at zero. Differentiating under the integral, with  $n = h_y/|h_y|$ , gives  $V_y = |h_y|$  and  $V_x = \int_0^y n \cdot h_{xy} ds$ , so  $|V_x| \leq C|y|$ . Both are continuous, and  $V_y > 0$  off zero. In the new coordinates  $f_X = (1, q)$  and  $f_V = (0, n)$ , where  $q = h_x - V_x n$ , so

$$G = \begin{pmatrix} 1 + |q|^2 & q \cdot n \\ q \cdot n & 1 \end{pmatrix}, \quad \det G = 1 + |q|^2 - (q \cdot n)^2 \geq 1.$$

Since  $|q| \leq CR$  and  $R^2 \leq C\rho$ , the stated bounds follow.

For the radii, Taylor's formula gives  $|h - wy^2| \leq C(x^2 + |xy| + |y|R^2)$ . Integrating  $||h_y(x, s)| - 2|w||s|| \leq C(|x| + x^2 + s^2)$  gives  $|V - \operatorname{sgn}(y)|w|y^2| \leq C(|xy| + |y|R^2)$ . Hence  $||h| - |V|| \leq CR\rho$ , and comparing the norms of  $(x, |h|)$  and  $(x, |V|)$  gives  $|\rho_e - \rho| \leq CR\rho \leq C\rho^{3/2}$ .  $\square$

**Corollary 3.3** (Intrinsic distance). *Let  $d_g$  be the path distance of  $g$  on the source. Near a cross-cap  $p$ ,*

$$d_g(\cdot, p) = \rho(1 + O(\rho^{1/2})), \quad \log d_g(\cdot, p) - \log \rho_e = O(\rho^{1/2}).$$

*Proof.* In the arclength coordinates,  $|v|^2(1 - C\rho^{1/2}) \leq v^T G v \leq |v|^2(1 + C\rho^{1/2})$ . The straight segment to a point at radius  $\rho$  has length at most  $\rho(1 + C\rho^{1/2})$ . A curve from  $p$  to that point either stays in the disk of radius  $2\rho$ , and then has length at least  $\rho(1 - C\rho^{1/2})$ , or crosses the annulus  $\rho \leq |\cdot| \leq 2\rho$ , which costs at least  $\rho(1 - C\rho^{1/2})$ . Lemma 3.2 compares  $\rho$  with  $\rho_e$ .  $\square$

*Remark 3.4.* Grieser proved that the Whitney umbrella metric is quasi-isometric to the Euclidean plane through a singular coordinate change [3, §1]. Lemma 3.2 sharpens this from bounded distortion to  $G = I + O(\rho^{1/2})$ . Bounded distortion would suffice for continuity of point values, but the exact constant  $-1/(2\pi)$  requires the sharper form.

## 4 The Friedrichs Laplacian

**Proposition 4.1** (Zero capacity). *Each interior cross-cap has zero capacity for  $q$ . Consequently  $\mathcal{V}$  is also the closure of  $C_c^\infty(\operatorname{int} \Sigma)$  in case (D), or of  $C^\infty(\Sigma)$  in case (N).*

*Proof.* In the arclength coordinates,  $q$  and the  $L^2$  norm on a small disk are equivalent to the Euclidean  $H^1$  and  $L^2$  norms. Cutoffs equal to  $\log(\varepsilon/\rho)/\log(1/\varepsilon)$  for  $\varepsilon^2 \leq \rho \leq \varepsilon$ , suitably smoothed, have Euclidean Dirichlet energy  $O(1/\log(1/\varepsilon))$ . Multiplying a smooth function by one minus such a cutoff therefore approximates it in the form norm by functions vanishing near the point. Smooth source functions have finite energy, because  $|\nabla u|_g^2 dA_g = O(R^{-1}) dx dy$  by Lemma 3.1.  $\square$

**Theorem 4.2** (Compactness). *The embedding  $\mathcal{V} \hookrightarrow \mathcal{H}$  is compact, so  $H_F$  has compact resolvent. In case (D),  $H_F \geq \lambda_D > 0$ . In case (N),  $\ker H_F$  consists of the constants.*

*Proof.* Fix  $p \in Z \cup Z_\partial$  and chart coordinates of Definition 2.1, in which a neighbourhood of  $p$  in  $\Sigma$  is a sector. Let  $\chi$  be a chart cutoff, equal to 1 near  $p$ , and let  $u \in \mathcal{C}$ , so that  $h = \chi u$  vanishes near  $p$ . On each ray, integrating  $(R|h|^2)'$  and applying Cauchy–Schwarz gives

$$\int_0^{R_0} |h|^2 dR \leq 4 \int_0^{R_0} R^2 |\partial_R h|^2 dR.$$

Integrating over the angular interval and using Lemma 3.1 ( $dA_g \asymp R dx dy$  and  $g \leq CI$ ),

$$\int \frac{|h|^2}{R^2} dA_g \leq C \int R |\nabla h|^2 dx dy \leq C'(q(u, u) + \|u\|^2).$$

This extends to  $\mathcal{V}$  by closure. Hence the  $L^2$  mass in  $\{R < \varepsilon\}$  is at most  $C\varepsilon^2$  times the form norm, uniformly over the finitely many singular points. On the complement of these small sectors,  $g$  is

a smooth Riemannian metric on a compact Lipschitz domain, where Rellich's theorem applies. A diagonal argument together with the uniform small-sector bound gives compactness.

If  $q(u_n, u_n) \rightarrow 0$  with  $\|u_n\| = 1$ , a subsequence converges in  $\mathcal{H}$  to a function with zero gradient on the connected regular locus, which is a constant. In case (D) its Dirichlet trace on a regular boundary arc vanishes, so the constant is zero, a contradiction. In case (N) the constants lie in  $\mathcal{V}$  and form the kernel.  $\square$

## 5 Point values and the extension problem

**Theorem 5.1** (Point traces). *Each  $u \in \text{Dom } H_F$  has a unique representative that is continuous at every  $p_j$ , Hölder continuous in the arclength coordinates. The map  $\tau u = (u(p_1), \dots, u(p_k))$  satisfies  $|\tau u| \leq C(\|u\| + \|H_F u\|)$ , is onto  $\mathbb{C}^k$ , and has kernel dense in  $\mathcal{H}$ .*

*Proof.* Near  $p_j$ , in the arclength coordinates,  $u$  is a local  $H^1$  weak solution of  $-\text{div}(a\nabla u) = m H_F u$ . This equation holds across the point because, by Proposition 4.1, test functions vanishing near the point are dense in  $H^1$ . The coefficients are real, bounded and uniformly elliptic, and  $m H_F u \in L^2$  with  $2 > \dim / 2$ , so the interior Hölder estimate [8, Thm. 8.24], [9, Lect. 18] gives  $\|u\|_{C^\alpha} \leq C(\|u\|_{L^2} + \|H_F u\|_{L^2})$  on a smaller disk. A cutoff  $\chi_j$  equal to 1 near  $p_j$ , supported away from the other singular points and from  $\partial\Sigma$ , has  $\Delta_g \chi_j$  smooth and supported on a regular annulus, so  $\chi_j \in \text{Dom } H_F$  and  $\tau \chi_j = e_j$ . Finally the core  $\mathcal{C}$  consists of smooth functions vanishing near  $Z$  (and near  $\partial\Sigma$  in case (D)); integration by parts on the regular locus puts it in  $\text{Dom } H_F \cap \ker \tau$ , and it is dense in  $\mathcal{H}$ .  $\square$

**Theorem 5.2** (The point-restricted Laplacian). *The operator  $A = H_F|_{\ker \tau}$  is closed, densely defined and symmetric, with deficiency indices  $(k, k)$ , and its Friedrichs extension is  $H_F$ . Fix  $\eta < 0$ , let  $R_z = (H_F - z)^{-1}$ , and put  $G_z = (\tau R_z)^* : \mathbb{C}^k \rightarrow \mathcal{H}$ . Then*

$$\text{Dom } A^* = \text{Dom } H_F \dot{+} G_\eta \mathbb{C}^k, \quad A^*(u + G_\eta a) = H_F u + \eta G_\eta a.$$

*The maps  $\Gamma_0(u + G_\eta a) = a$  and  $\Gamma_1(u + G_\eta a) = \tau u$  form a boundary triple. The self-adjoint extensions are exactly the operators  $(U - I)\Gamma_1 + i(U + I)\Gamma_0 = 0$ ,  $U \in U(k)$ , and their resolvents differ from  $R_z$  by operators of rank at most  $k$ . All of them have compact resolvent. In case (D), the Krein-von Neumann extension has domain  $\text{Dom } A \dot{+} G_0 \mathbb{C}^k$  and kernel of dimension  $k$ .*

*Proof.* Graph continuity makes  $\ker \tau$  graph-closed. For nonreal  $z$ , the bounded surjection  $\tau R_z : \mathcal{H} \rightarrow \mathbb{C}^k$  has kernel  $\text{ran}(A - z)$ , so  $\dim \ker(A^* - \bar{z}) = k$ . The form core  $\mathcal{C}$  lies in  $\text{Dom } A$ , so the Friedrichs extension of  $A$  is  $H_F$ . The decomposition and the Green identity

$$\langle A^* \varphi, \psi \rangle - \langle \varphi, A^* \psi \rangle = \langle \Gamma_1 \varphi, \Gamma_0 \psi \rangle - \langle \Gamma_0 \varphi, \Gamma_1 \psi \rangle$$

follow from  $\langle (H_F - \eta)u, G_\eta b \rangle = \langle \tau u, b \rangle$ . This is the framework of [7]. Resolvent differences have range in  $\ker(A^* - z)$ , which has dimension  $k$ , and  $R_z$  is compact. In case (D)  $A$  is strictly positive, and the Krein theorem [10, Thm. 2.1] gives  $\text{Dom } A_K = \text{Dom } A \dot{+} \ker A^*$ , with  $\ker A^* = G_0 \mathbb{C}^k$  since  $0 \in \rho(H_F)$ .  $\square$

## 6 Logarithmic normalization

**Theorem 6.1** (Green vectors). *For real  $\eta < 0$  there are  $\beta > 0$  and a real symmetric matrix  $B_\eta$  such that, near each  $p_i$ ,*

$$G_{\eta,j} = -\frac{\delta_{ij}}{2\pi} \log d_g(\cdot, p_i) + (B_\eta)_{ij} + O(d_g(\cdot, p_i)^\beta).$$

*The same holds with  $d_g$  replaced by the extrinsic radius  $\rho_{e,i} = |f - f(p_i)|$ , with the same matrix  $B_\eta$ .*

*Proof.* In arclength coordinates at  $p_i$ , let  $h_0 = -\chi \log \rho / (2\pi)$  with a cutoff  $\chi$ . Then  $-\Delta h_0 = \delta_0 + k_0$  with  $k_0$  bounded and supported in the cutoff annulus. Writing  $E = a - I$ ,

$$(-\operatorname{div}(a\nabla) - \eta m)h_0 = \delta_0 + \mathcal{F},$$

where  $\mathcal{F}$  consists of a divergence term with density  $E\nabla h_0$  and a scalar term  $k_0 - \eta m h_0$ . Since  $|E\nabla h_0| \leq C\rho^{-1/2}$ , the divergence density lies in  $L^3$ , and the scalar term lies in  $L^2$ . Thus  $\mathcal{F}$  is a bounded functional on  $\mathcal{V}$ . Lax–Milgram for the coercive form  $q - \eta\langle \cdot, \cdot \rangle$  gives  $w \in \mathcal{V}$  with  $q_\eta(w, \cdot) = \mathcal{F}$ , and the Hölder estimate with divergence data in  $L^q$ ,  $q > 2$  [9, Lect. 18], makes  $w$  Hölder continuous. So  $g_i = h_0 - w$  solves the point-source equation, with a logarithm plus a Hölder regular part.

To identify  $g_i$  with  $G_{\eta,i}$ , take  $u \in \operatorname{Dom} H_F$ . The Hölder estimate gives  $|u - u(p_i)| \leq C\rho^\alpha$ , and testing the equation with a cutoff times  $u - u(p_i)$  gives  $\int_{B_r} |\nabla u|^2 \leq Cr^{2\alpha}$ . The improper pairing  $\int a\nabla h_0 \cdot \nabla u$  therefore converges. Excising  $B_\varepsilon$  with cutoffs of gradient  $O(1/\varepsilon)$  costs  $O(|\log \varepsilon| \varepsilon^\alpha)$ , and the flux of  $-\log \rho / (2\pi)$  contributes  $u(p_i)$  in the limit. Hence  $\langle g_i, (H_F - \eta)u \rangle = u(p_i)$  for all  $u$ , which is the defining property of  $G_{\eta,i} = (\tau_i R_\eta)^* 1$ . The coefficients are real since  $q$  and  $\eta$  are. Symmetry follows from self-adjointness of  $R_\eta$  applied to normalized indicators shrinking to the poles, using uniform convergence near the other poles. Corollary 3.3 and Lemma 3.2 replace  $\log \rho$  by  $\log d_g$  or  $\log \rho_e$ , at the cost of reducing  $\beta$  to at most  $1/2$ .  $\square$

**Theorem 6.2** (Geometric boundary coordinates). *Every  $\varphi \in \operatorname{Dom} A^*$  has unique coefficients  $\ell, b \in \mathbb{C}^k$  with*

$$\varphi = \ell_i \log d_g(\cdot, p_i) + b_i + O(d_g^\beta) \quad \text{at } p_i,$$

*namely  $\ell = -a/(2\pi)$  and  $b = \tau u + B_\eta a$  for  $\varphi = u + G_\eta a$ . The map  $\varphi \mapsto (\ell, b)$  is graph-bounded, onto  $\mathbb{C}^k \oplus \mathbb{C}^k$ , and has kernel  $\operatorname{Dom} A$ . If  $\psi$  has coefficients  $(\ell', b')$ , then*

$$\langle A^* \varphi, \psi \rangle - \langle \varphi, A^* \psi \rangle = 2\pi(\langle \ell, b' \rangle - \langle b, \ell' \rangle).$$

*The Friedrichs domain is  $\{\ell = 0\}$ , and the self-adjoint extensions are the maximal isotropic planes of this form, namely  $(U - I)b + i(U + I)a = 0$  with  $a = -2\pi\ell$  and  $U \in U(k)$ . For a Hermitian matrix  $C$ , the extension  $b = Ca$  has resolvent  $R_z + G_z(C - B_\eta - M_z)^{-1}G_z^*$ , where  $M_z = (z - \eta)\tau R_z G_\eta$ .*

*Proof.* Substitute Theorem 6.1 into the decomposition of Theorem 5.2. Uniqueness follows by dividing by the logarithm. The pairing changes from  $\langle \tau u, a' \rangle - \langle a, \tau u' \rangle$  by the symmetric shift  $B_\eta$ , which cancels. The resolvent formula follows from  $G_z - G_\eta = (z - \eta)R_z G_\eta \in \operatorname{Dom} H_F$ .  $\square$

## 7 Uniqueness, scale and sheets

**Corollary 7.1** (Markov uniqueness).  *$H_F$  is the only self-adjoint extension of  $A$  whose semigroup is sub-Markovian.*

*Proof.*  $q$  is a Dirichlet form, so  $H_F$  is sub-Markovian. If  $H$  is another sub-Markovian extension, then  $\|(H + \lambda)^{-1}h\|_\infty \leq \lambda^{-1}\|h\|_\infty$  for bounded  $h \in \mathcal{H}$ . By Theorem 6.2, a nonzero logarithmic coefficient would make  $(H + \lambda)^{-1}h$  unbounded near its pole, since the density is bounded below in arclength coordinates. All coefficients therefore vanish, the two resolvents agree on a dense set, and  $H = H_F$ .  $\square$

The same argument shows that, in the plane, the free Laplacian is the only Markovian point interaction with respect to Lebesgue measure. Dirichlet forms associated with point interactions are studied in [11].

**Corollary 7.2** (Reference length). *Replacing  $\log d_g$  by  $\log(d_g/\lambda_i)$  changes  $b$  to  $b + \operatorname{diag}(\log \lambda_i)\ell$ . The only self-adjoint boundary plane invariant under every common change  $\lambda_i \mapsto e^t \lambda_i$  is the Friedrichs plane  $\ell = 0$ .*

*Proof.* If  $(\ell, b)$  lies in an invariant plane, so does  $(0, \ell)$ . Pairing the two gives  $2\pi\|\ell\|^2 = 0$ .  $\square$

**Corollary 7.3** (Sheet-even coefficients). *If  $s_\varepsilon \neq t_\varepsilon$  tend to the same  $p_i$  with  $f(s_\varepsilon) = f(t_\varepsilon)$ , then for every  $\varphi \in \text{Dom } A^*$ ,  $\varphi(s_\varepsilon) - \varphi(t_\varepsilon) = O(\rho_\varepsilon^\beta) \rightarrow 0$ . The coefficients are the same on both sheets.*

*Proof.* The two points have the same extrinsic radius, so the expansions in  $\rho_{\varepsilon,i}$  agree up to the remainders.  $\square$

## 8 Why the boundary must be fixed

Let  $H_0$  be the graph closure of  $-\Delta_g$  on  $C_c^\infty(\text{int } \Sigma \setminus Z)$  in case (D).

**Proposition 8.1.** *The deficiency spaces of  $H_0$  are infinite-dimensional, and  $H_0$  has at least two distinct sub-Markovian self-adjoint extensions.*

*Proof.* Take a regular boundary arc. For smooth boundary data  $\phi$  supported in it, let  $w_\phi$  be a smooth collar extension, supported away from  $Z \cup Z_\partial$ , and put  $v_\phi = w_\phi - (H_F - z)^{-1}(-\Delta_g - z)w_\phi$  for nonreal  $z$ . Then  $(-\Delta_g - z)v_\phi = 0$  in the interior, so  $v_\phi \in \ker(H_0^* - z)$ , and the trace of  $v_\phi$  on the arc is  $\phi$ . Independent  $\phi$  give independent  $v_\phi$ . For the second claim,  $H_F$  is one sub-Markovian extension. Closing  $q$  on smooth functions with free boundary values gives another, the Neumann Laplacian, whose kernel contains the constants.  $\square$

The finite classification of Section 5 therefore requires the outer boundary condition to be part of the definition of the operator.

## 9 Examples

### 9.1 Steiner's Roman surface

Let  $\Sigma = \mathbb{RP}^2$  and let  $f[x : y : z] = (yz, zx, xy)$  on unit representatives. This is case (N).

**Proposition 9.1.**  *$f$  is an immersion except at the six points  $[1 : \pm 1 : 0]$ ,  $[1 : 0 : \pm 1]$  and  $[0 : 1 : \pm 1]$ , each of which is a Whitney cross-cap. In an orthonormal adapted frame, the Whitney determinant (1) equals  $\pm 2$ .*

*Proof.* On  $S^2$  the differential is the restriction of  $M = \begin{pmatrix} 0 & z & y \\ z & 0 & x \\ y & x & 0 \end{pmatrix}$ , and  $\det M = 2xyz$ . The restriction to the tangent plane has rank below two only if  $M$  has a kernel vector tangent to the sphere. This forces one coordinate to vanish and the other two to be equal up to sign, which gives twelve points on  $S^2$  and six on  $\mathbb{RP}^2$ . With  $\eta$  the tangent kernel vector,  $\xi = p \times \eta$  and the chart  $(s, t) \mapsto (p + s\xi + t\eta)/|p + s\xi + t\eta|$ , an exact computation (replayed in the verifier) gives  $f_t = 0$ ,  $|f_s| = 1$  and  $\det(f_s, f_{st}, f_{tt}) = \pm 2$ .  $\square$

Theorems 5.1–6.2 therefore give a  $U(6)$ -family of pseudo-Laplacians on the Roman surface, each with six logarithmic channels.

**Theorem 9.2** (Symmetry of the Green matrix). *Index the cross-caps so that  $P_\perp$  pairs each with the unique one whose line is orthogonal to it, and let  $P_\circ = J - I - P_\perp$  be the adjacency of the remaining pairs, whose lines meet at  $60^\circ$ . For every  $\eta < 0$  there are real numbers  $r, \gamma_\perp, \gamma_\circ$  with*

$$B_\eta = rI + \gamma_\perp P_\perp + \gamma_\circ P_\circ.$$

*Hence  $B_\eta$  has at most three eigenvalues:  $r + \gamma_\perp + 4\gamma_\circ$  on the constants,  $r + \gamma_\perp - 2\gamma_\circ$  with multiplicity two, and  $r - \gamma_\perp$  with multiplicity three.*

*Proof.* Each signed permutation  $\sigma$  of coordinates descends to an isometry of  $(\mathbb{RP}^2, g)$ , because  $f \circ \sigma = L_\sigma \circ f$  for a signed permutation  $L_\sigma$  of  $\mathbb{R}^3$ , which is a Euclidean isometry. The pullback by  $\sigma$  is unitary, commutes with  $H_F$  and  $R_\eta$ , permutes the cross-caps, and preserves extrinsic radii. Hence it permutes the Green vectors, and  $(B_\eta)_{\sigma i \sigma j} = (B_\eta)_{ij}$ . This group acts transitively on the six cross-caps, and its orbits on ordered pairs of distinct cross-caps are exactly the six orthogonal pairs and the twenty-four  $60^\circ$  pairs (verified). Symmetry of  $B_\eta$  completes the form.  $P_\perp$  and  $P_\circ$  commute;  $P_\circ$  is the adjacency matrix of the octahedron. Their joint eigenvalues are  $(1, 4)$  on the constants,  $(1, -2)$  on matching-even vectors of sum zero, and  $(-1, 0)$  on matching-odd vectors.  $\square$

## 9.2 A ruled surface with boundary cross-caps

The ruled map

$$S(t, u) = (\cos t + 2u(1 - \cos t), (1 - u) \sin t, u\sqrt{\cos t(2 - \cos t)})$$

on  $[-\pi/2, \pi/2] \times [0, 1]$ , completed at  $t = \pm\pi/2$  by the chart  $r = Q$ , arose in a two-component signal model [13, 12]. It has five rank-one points, all Whitney cross-caps [12]. Two are interior:  $P_\pm = (\pm t_0, u_0)$  with  $\cos t_0 = (3 - \sqrt{5})/2$  and  $u_0 = (1 - \cos t_0)/2$ . Three lie on the boundary: a half-germ at  $(0, 1)$  and two quarter-germs at the lower corners. In the completed charts the boundary is formed by coordinate lines, so Definition 2.1 holds in case (D) with  $k = 2$ . All results apply. Moreover  $S(-t, u) = JS(t, u)$  with  $J = \text{diag}(1, -1, 1)$ , and the argument of Theorem 9.2 shows  $B_\eta = \begin{pmatrix} r & \gamma \\ \gamma & r \end{pmatrix}$ . The Krein extension has a two-dimensional kernel. An earlier preprint on this surface [14] defined the two-point problem through the compact-core closure  $H_0$  and claimed indices  $(2, 2)$ ; Proposition 8.1 shows why the outer boundary must be fixed first.

## 10 Open problems

- *Explicit regular parts.* Compute or estimate  $B_\eta$ , for instance the three Roman-surface parameters, and relate them to global geometry.
- *Spectral asymptotics.* The finite-rank structure transfers eigenvalue counting from  $H_F$  to every extension. A Weyl law for  $H_F$  at degenerate metrics, and heat-trace contributions of cross-caps, remain to be determined.
- *Dirac operators.* A spinor or twisted Dirac analogue at cross-caps needs closed domains, cut traces and transmission conditions. The square-root monodromy of the observation field of the ruled surface suggests a natural twist [13].
- *Other singularities.* Non-generic germs such as cuspidal edges or  $S_k^\pm$  singularities may carry different channel structures.

## A Evidence

The proofs above are written in full. External inputs:

- the cross-cap recognition criterion [6] and Whitney’s genericity theorem [4, 5];
- the scalar Hölder estimates [8, 9];
- the abstract extension and Krein theorems [7, 10].

An exact replay checks:

- the adapted metric matrix and determinant identities;

- the Roman-surface singular set and Whitney determinants;
- the symmetry orbits on pairs of cross-caps;
- the eigenvalue pattern of  $rI + \gamma_{\perp}P_{\perp} + \gamma_{\circ}P_{\circ}$ ;
- the symmetry identity of the ruled surface.

These checks confirm the algebra only, not the analysis. No Lean formalization of the analytic statements is claimed.

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